



A finite-difference method for the numerical solution of the Schrödinger equation

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Abstract

A new approach, which is based on a new property of phase-lag for computing eigenvalues of Schrödinger equations with potentials, is developed in this paper. We investigate two cases: (i) The specific case in which the potential $V(x)$ is an even function with respect to x . It is assumed, also, that the wave functions tend to zero for $x \rightarrow \pm\infty$. (ii) The general case of the Morse potential and of the Woods–Saxon or optical potential. Numerical and theoretical results show that this new approach is more efficient compared to previously derived methods.

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1. Introduction

In recent years the Schrödinger equation has been the subject of great activity, the aim being to achieve a fast and reliable algorithm that generates a numerical solution (see [4, 7, 12–18, 27–35, 23, 3, 37–39]).

The one dimensional Schrödinger equation has the form

$$y''(x) = [V(x) - E]y(x). \quad (1)$$

Equations of this type occur very frequently in theoretical physics, for example [26], and there is a real need to be able to solve them both efficiently and reliably by numerical methods. In (1), E is a real number denoting *the energy* and V is a given function which denotes the potential. We investigate two cases.

In the first case, $V(x)$ is an even function and $y(x) \rightarrow 0$ for $x \rightarrow \pm\infty$. As examples of potentials which satisfy these properties we present the following potentials, which are well known in several areas of physics:

(i) The one-dimensional anharmonic oscillator potential,

$$V_i(x) = x^2 + \frac{\lambda x^2}{1 + \gamma x^2} \quad \text{with } \lambda \text{ and } \gamma \text{ parameters.} \quad (2)$$

(ii) The symmetric double-well potential,

$$V_{ii}(x) = x^6 - \beta x^2 \quad \text{with } \beta \text{ a parameter.} \quad (3)$$

(iii) The Razavy potential,

$$V_{iii}(x) = \frac{1}{8}m^2(\cosh(4x) - 1) - m(n+1)\cosh(2x) \quad \text{with } n \text{ and } m \text{ parameters.} \quad (4)$$

For the numerical solution of the specific eigenvalue Schrödinger equation (1) the following procedures have been obtained: (1) Rayleigh–Ritz methods (see [27]), perturbation methods (see [4]), methods using Padé approximants (see [23]), direct numerical integration techniques or boundary value techniques (see [13–15, 31]) and an operator method based upon the $SO(2, 1)$ dynamic group (see [13]). We note here that the method based on $SO(2, 1)$ dynamic group gives much more accurate results than all finite-difference methods used. Analytical approaches to the Schrödinger equation (1) have been obtained for $V_i(x)$ given by (2) by Flessas [17, 18], Varma [38], Whitehead et al. [39] for $V_{ii}(x)$ given by (4) by Razavy [30]. We must note here that in these cases the matrix methods are more useful than the shooting techniques.

Fack and Vanden Berghe [16] have shown numerically that their method is the most accurate direct finite-difference method for the numerical solution of the eigenvalue problem (1).

In the second case $V(x)$ is a general function. As examples, we present the following potentials, which are well known in several areas of physics:

(i) The Morse potential (see [1, 28]):

$$V_{gi}(x) = Dt(t - 2), \quad t = \exp(aX), \quad (5)$$

where $X = x_e - x$, $x_e = 1.9975$, $a = 0.711248$, and $D = 188.4355$.

(ii) The well-known Woods–Saxon potential (see [1]):

$$V_{gii}(x) = \frac{u_0}{1 + t} - \frac{u_0 t}{a_0(1 + t)^2}, \quad (6)$$

where $t = \exp((x - x_e)/a_0)$, $u_0 = -50$, $x_e = 7$ and $a_0 = 0.6$.

For the numerical computation of the eigenvalues of the Morse potential (5) and the Woods–Saxon potential (6) there are, also, other numerical methods which give very accurate results (see [37, 20–22]). These methods are based on the shooting technique, which is completely different from the technique used by the matrix methods.

In Section 2 we will develop the basic theory for the phase-lag analysis of the symmetric four-step methods. A simple four-step method with minimal phase-lag has been constructed in Section 3. Finally, in Section 4 the application of the developed method to the problem (1) has been presented and extended numerical results based on the potentials V_i , V_{ii} , V_{iii} , V_{gi} and V_{gii} are produced to show the efficiency of the new approach.

2. Phase-lag analysis of general symmetric $2k$ -step, $k \in N$ methods

In recent years there has been considerable interest in the numerical solution of second-order periodic initial-value problems (see [6, 8–11, 25, 29, 36]):

$$y'' = f(x, y), \quad y(x_0) = y_0, \quad y'(x_0) = y'_0. \quad (7)$$

To investigate the absolute stability properties we introduce the scalar test equation

$$y'' = -w^2 y. \quad (8)$$

When we apply a symmetric $2k$ method to the scalar test equation (8) we obtain a difference equation of the form

$$A_k(H)y_{n+k} + \cdots + A_1(H)y_{n+1} + A_0(H)y_n + A_1(H)y_{n-1} + \cdots + A_k(H)y_{n-k} = 0, \quad (9)$$

where $H = wh$, h is the step length and $A_0(H), A_1(H), \dots, A_k(H)$ are polynomials of H and y_n is the computed approximation to $y(nh)$, $n = 0, 1, 2, \dots$

The characteristic equation associated with (9) is

$$A_k(H)s^k + \cdots + A_1(H)s + A_0(H) + A_1(H)s^{-1} + \cdots + A_k(H)s^{-k} = 0. \quad (10)$$

Based on Lambert and Watson [25] we have the following definition.

Definition 1. A symmetric $2k$ -step method with the characteristic equation given by (10) is said to have an *interval of periodicity* $[H_0, H_1]$ if, for all $H \in [H_0, H_1]$, the roots s_i , $i = 1, \dots, 2k$ of (10) satisfy

$$s_1 = e^{i\theta(H)}, \quad s_2 = e^{-i\theta(H)}, \quad |s_i| \leq 1, \quad i = 3, \dots, 2k, \quad (11)$$

where $\theta(H)$ is a real function of H .

Definition 2. For any method corresponding to the characteristic equation (10) the *phase-lag* is defined as the leading term in the expansion of

$$t = H - \theta(H). \quad (12)$$

Then if the quantity $t = O(H^{q+1})$ as $H \rightarrow 0$, the order of phase-lag is q .

The phase-lag theory developed in the paper is valid for the case where w is imaginary as well. In that case we have the *exponential error* which is equivalent to *phase-lag*.

Theorem 1. For all H in the interval of periodicity, the symmetric $2k$ -step method with characteristic equation given by (10) has phase-lag order q and phase-lag constant c given by

$$-cH^{q+2} + O(H^{q+3}) = \frac{2A_k(H)\cos(kH) + \cdots + 2A_j(H)\cos(jH) + \cdots + A_0(H)}{2k^2A_k(H) + \cdots + 2j^2A_j(H) + \cdots + 2A_1(H)}. \quad (13)$$

Proof. If we put $s = e^{i\theta(H)}$ then (10) becomes

$$2A_k(H) \cos(k\theta(H)) + \cdots + 2A_j(H) \cos(j\theta(H)) + \cdots + A_0(H) = 0. \quad (14)$$

By definition the phase-lag order q and phase-lag constant c are given by

$$t = H - \cos(\theta(H)) = cH^{q+1} + O(H^{q+2}). \quad (15)$$

Then since

$$\theta(H) = H - t, \quad (16)$$

we may show from trigonometric expansions that

$$\cos(\theta(H)) = \cos(H - t) = \cos H + cH^{q+2} + O(H^{q+3}), \quad (17)$$

$$\sin(\theta(H)) = \sin(H - t) = \sin H - cH^{q+1} + O(H^{q+2}). \quad (18)$$

By an inductive argument using the familiar identities

$$\cos(j\theta(H)) = \cos((j-1)\theta(H)) \cos(\theta(H)) - \sin((j-1)\theta(H)) \sin(\theta(H)), \quad (19)$$

$$\sin(j\theta(H)) = \sin((j-1)\theta(H)) \cos(\theta(H)) + \cos((j-1)\theta(H)) \sin(\theta(H)). \quad (20)$$

It is now straightforward to show that

$$\cos(j\theta(H)) = \cos(jH) + cj^2H^{q+2} + O(H^{q+3}), \quad (21)$$

$$\sin(j\theta(H)) = \sin(jH) - cjH^{q+1} + O(H^{q+2}). \quad (22)$$

Then substituting (21) and (22) into (14) for $j = 1, 2, \dots, k$ will give the result.

The converse of the theorem may also be easily shown. The converse states that if (13) is true then the method will have phase-lag order q and phase-lag constant c . To prove this we suppose that the method has phase-lag order p and phase-lag constant d . This means

$$t = H - \theta(H) = dH^{p+1} + O(H^{p+1}). \quad (23)$$

Then we may show by trigonometric expansions that

$$\cos(j\theta(H)) = \cos(jH) + dj^2H^{p+2} + O(H^{p+3}). \quad (24)$$

Substituting for $\cos(jH)$ in (13) and using (14) we have

$$-2dH^{p+1} \sum_{j=0}^k A_j j^2 = -2cH^{q+1} \sum_{j=0}^k A_j j^2 + O(H^{q+2}). \quad (25)$$

Equating highest powers gives

$$p = q \quad \text{and} \quad d = c, \quad (26)$$

i.e., the theorem is proved. $\square$

The formula proposed from the above theorem gives us a direct method to calculate the phase-lag of any symmetric $2k$ -step method. It is obvious that for the symmetric four-step methods the phase-lag order and the phase-lag constant are given by (13) with $k = 2$.

The formula for the calculation of the exponential error is exactly the same as the formula for the calculation of the phase-lag, i.e., formula (13) but for w imaginary the $\cos$ and $\sin$ are replaced by $\cosh$ and $\sinh$.

3. Derivation of the new method

Based on the method developed by Henrici [19], we introduce the following four-step method (with one free parameter):

$$\begin{aligned} \bar{y}_n &= y_n - ah^2(y''_{n+2} - 4y''_{n+1} + 6y''_n - 4y''_{n-1} + y''_{n-2}), \\ y_{n+2} - 2y_{n+1} + 2y_n - 2y_{n-1} + y_{n-2} &= \frac{1}{120}h^2(9y''_{n+2} + 104y''_{n+1} + 14\bar{y}_n'' + 104y''_{n-1} + 9y''_{n-2}), \end{aligned} \quad (27)$$

where $y''_{n+2} = f(x_{n+2}, y_{n+2})$, $y''_{n+1} = f(x_{n+1}, y_{n+1})$, $y''_n = f(x_n, y_n)$, $y''_{n-1} = f(x_{n-1}, y_{n-1})$, $y''_{n-2} = f(x_{n-2}, y_{n-2})$, $\bar{y}_n'' = f(x_n, \bar{y}_n)$ and a is a constant to be suitably chosen.

Applying the Taylor series expansions of y_{n+2} , y_{n+1} , y_n , y_{n-1} , y_{n-2} , y''_{n+2} , y''_{n+1} , y''_n , y''_{n-1} and y''_{n-2} in (27) we have the following result for the local truncation error (LTE) of the method (27):

$$\text{LTE} = \frac{h^8}{30\,240}[-95y_n^{(8)} + 3528ay_n^{(6)}F_n] + O(h^{10}), \quad (28)$$

where $F = \partial f / \partial x$.

Applying the new method (27) to the scalar test equation (8) we have the difference scheme (9) and the associated characteristic equation (10) with

$$\begin{aligned} A_2(H) &= 1 + \frac{3}{40}H^2 + \frac{7}{60}aH^4, \\ A_1(H) &= -2 + \frac{13}{15}H^2 - \frac{7}{15}aH^4, \\ A_0(H) &= 2 + \frac{7}{60}H^2 + \frac{7}{10}aH^4. \end{aligned} \quad (29)$$

Applying (13) in Theorem 1 to (29) (with $k=2$) and expanding $\cos(2H)$ and $\cos(H)$ via Taylor series, we have the following expression for the phase-lag:

$$t = \frac{H^7(3528a - 95)}{120\,960} + \frac{H^9(4200a - 121)}{345\,600} + O(H^{11}). \quad (30)$$

To have minimal phase-lag

$$a = \frac{95}{3528}, \quad (31)$$

and for this a the phase-lag is given by

$$t = -\frac{83H^9}{3\,628\,800} + O(H^{11}). \quad (32)$$

The above analysis lead us to the following theorem.

Theorem 2. The method (27) with a given by (31) is a sixth-order method with phase-lag of order eight.

The result produced for minimal exponential error is exactly the same as that produced for minimal phase-lag. The parameter a is the same as in (31) and we recover Eq. (32).

In Appendix A we present the phase-lag analysis of the method of Fack et al. [16]. In Appendix B we investigate the interval of periodicity of the present method and of the method of Fack et al. [16].

4. Numerical illustrations

4.1. Computer implementation

Case i: We must note that although the solutions of (1) are defined in the interval $(-\infty, +\infty)$, these solutions are either of even or odd parity, i.e., we have $y(x) = \pm y(-x)$. So, the numerical solution of (1) can be restricted into the region $[0, +\infty)$. Furthermore, it is assumed that the wave functions satisfy the Dirichlet boundary condition $y(x) = 0$ at some $x = R$, to which a value of R is specified. For the numerical integration of (1) the interval $[0, R]$ is divided into N parts of length $h = R/N$, such that $y_0 = y(0)$, $y_1 = y(h)$, ..., $y_N = y(Nh) = y(R)$.

Case ii: For the second case we have the numerical integration of (1) over the interval $[-R, R]$. This interval is divided into N parts of length $h = 2R/N$, such that $y_0 = y(0)$, $y_1 = y(h)$, ..., $y_N = y(Nh) = y(R)$.

Assuming that $y_n = y(nh)$ and applying (27) to (1) we have the following difference equation:

$$\begin{aligned} & \frac{120 - 9h^2V_{n+2} + 14ah^4V_nV_{n+2}}{120}y_{n+2} - \frac{30 + 13h^2V_{n+1} + 7ah^4V_nV_{n+1}}{15}y_{n+1} \\ & + \frac{120 - 7h^2V_n + 42ah^4V_n^2}{60}y_n - \frac{30 + 13h^2V_{n-1} + 7ah^4V_nV_{n-1}}{15}y_{n-1} \\ & + \frac{120 - 9h^2V_{n-2} + 14ah^4V_nV_{n-2}}{120}y_{n-2} \\ & = -h^2E \left[\frac{9 - 14ah^2(V_n + V_{n+2})}{120}y_{n+2} + \frac{7ah^2(V_n + V_{n+1}) + 13}{15}y_{n+1} \right. \\ & \quad \left. + \frac{7(1 - 12ah^2V_n)}{60}y_n + \frac{7ah^2(V_n + V_{n-1}) + 13}{15}y_{n-1} + \frac{9 - 14ah^2(V_n + V_{n-2})}{120}y_{n-2} \right] \\ & \quad - \frac{7ah^4E^2}{60}(y_{n+2} - 4y_{n+1} + 6y_n - 4y_{n-1} + y_{n-2}), \end{aligned} \quad (33)$$

where $V_k = V(kh)$, $k = n+2, n+1, n, n-1, n-2$.

Considering that we have even or odd parity wave functions, i.e., $y_k = \pm y_{-k}$, it is easy to obtain the following discretization of (1):

$$AY = -h^2E^{\text{calc}}BY + h^4(E^{\text{calc}})^2CY \quad (34)$$

with matrices A, B , and C having the following pentadiagonal form:

$$(a_{1,1}, a_{1,2}, a_{1,3}) = \begin{cases} (S_{1,1}, S_{1,2}, S_{1,3}) & \text{for even parity solutions,} \\ (S_{1,1}, 0, 0) & \text{for odd parity solutions,} \end{cases} \quad (35)$$

with

$$\begin{aligned} S_{1,1} &= \frac{120 - 7h^2 V_0 + 42ah^4 V_0^2}{60}, \\ S_{1,2} &= -2 \frac{7ah^4 V_0 V_1 + 13h^2 V_1 + 30}{15}, \\ S_{1,3} &= \frac{-9h^2 V_2 + 14ah^4 V_0 V_2 + 12}{60}; \\ (b_{1,1}, b_{1,2}, b_{1,3}) &= \begin{cases} (Q_{1,1}, Q_{1,2}, Q_{1,3}) & \text{for even parity solutions,} \\ (Q_{1,1}, 0, 0) & \text{for odd parity solutions,} \end{cases} \end{aligned} \quad (36)$$

where

$$\begin{aligned} Q_{1,1} &= \frac{7(1 - 12ah^2 V_0)}{60}, \\ Q_{1,2} &= 2 \frac{7ah^2(V_0 + V_1) + 13}{15}, \\ Q_{1,3} &= \frac{9 - 14ah^2(V_0 + V_2)}{60}; \\ (c_{1,1}, c_{1,2}, c_{1,3}) &= \begin{cases} (-\frac{7}{10}a, \frac{14}{15}a, -\frac{7}{30}a) & \text{for even parity solutions,} \\ (-\frac{7}{10}a, 0, 0) & \text{for odd parity solutions,} \end{cases} \end{aligned} \quad (37)$$

$$(a_{2,1}, a_{2,2}, a_{2,3}, a_{2,4}) = (S_{2,1}, S_{2,2}, S_{2,3}, S_{2,4}), \quad (38)$$

where

$$\begin{aligned} S_{2,1} &= -\frac{7ah^4 V_0 V_1 + 13h^2 V_0 + 30}{15}, \\ S_{2,2} &= \frac{120 - 7h^2 V_1 + 42ah^4 V_1^2}{60} \pm \frac{-9h^2 V_1 + 14ah^4 V_1^2 + 120}{120}, \\ S_{2,3} &= -\frac{7ah^4 V_1 V_2 + 13h^2 V_2 + 30}{15}, \\ S_{2,4} &= \frac{-9h^2 V_3 + 14ah^4 V_1 V_3 + 120}{120}; \\ (b_{2,1}, b_{2,2}, b_{2,3}, b_{2,4}) &= (T_1, T_2, T_3, T_4), \end{aligned} \quad (39)$$

where

$$T_1 = \frac{7ah^2(V_0 + V_1) + 13}{15},$$

$$\begin{aligned}
T_2 &= \frac{7h^2(1 - 12ah^2V_1)}{60} \pm h^2 \frac{9 - 28ah^2V_1}{120}, \\
T_3 &= \frac{7ah^2(V_1 + V_2) + 13}{15}, \\
T_4 &= \frac{9 - 14ah^2(V_1 + V_3)}{120} : \\
(c_{2,1}, c_{2,2}, c_{2,3}, c_{2,4}) &= \left(\frac{7}{15}a, -\frac{7}{10}a \pm \frac{7}{60}a, \frac{7}{15}a, -\frac{7}{60}a\right)
\end{aligned} \tag{40}$$

with the plus sign for even-parity wave functions and minus sign for odd parity wave functions:

$$(a_{n,n-2}, a_{n,n-1}, a_{n,n}, a_{n,n+1}, a_{n,n+2}) = (G_1, G_2, G_3, G_4, G_5), \tag{41}$$

where

$$\begin{aligned}
G_1 &= \frac{-9h^2V_{n-2} + 14ah^4V_nV_{n-2} + 120}{120}, \\
G_2 &= -\frac{7ah^4V_nV_{n-1} + 13h^2V_{n-1} + 30}{15}, \\
G_3 &= \frac{120 - 7h^2V_n + 42ah^4V_n^2}{60}, \\
G_4 &= -\frac{7ah^4V_nV_{n+1} + 13h^2V_{n+1} + 30}{15}, \\
G_5 &= \frac{-9h^2V_{n+2} + 14ah^4V_nV_{n+2} + 120}{120} : \\
(b_{n,n-2}, b_{n,n-1}, b_{n,n}, b_{n,n+1}, b_{n,n+2}) &= (H_1, H_2, H_3, H_4, H_{2,5}),
\end{aligned} \tag{42}$$

where

$$\begin{aligned}
H_1 &= \frac{9 - 14ah^2(V_n + V_{n-2})}{120}, \\
H_2 &= \frac{7ah^2(V_n + V_{n-1}) + 13}{15}, \\
H_3 &= \frac{7(1 - 12ah^2V_n)}{60}, \\
H_4 &= \frac{7ah^2(V_n + V_{n+1}) + 13}{15}, \\
H_5 &= \frac{9 - 14ah^2(V_n + V_{n+2})}{120} : \\
(c_{n,n-2}, c_{n,n-1}, c_{n,n}, c_{n,n+1}, c_{n,n+2}) &= \left(-\frac{7}{60}a, \frac{7}{15}a, -\frac{7}{10}a, \frac{7}{15}a, -\frac{7}{60}a\right)
\end{aligned} \tag{43}$$

for $n = 3(1)N - 2$:

$$(a_{N-1,N-3}, a_{N-1,N-2}, a_{N-1,N-1}, a_{N-1,N}) = (F_1, F_2, F_3, F_4), \quad (44)$$

where

$$\begin{aligned} F_1 &= \frac{-9h^2 V_{N-4} + 14ah^4 V_{N-2} V_{N-4} + 120}{120}, \\ F_2 &= -\frac{7ah^4 V_{N-2} V_{N-3} + 13h^2 V_{N-3} + 30}{15}, \\ F_3 &= \frac{120 - 7h^2 V_{N-2} + 42ah^4 V_{N-2}^2}{60} \pm \frac{-9h^2 V_{N-2} + 14ah^4 V_{N-2}^2 + 120}{120}, \\ F_4 &= -\frac{7ah^4 V_{N-2} V_{N-1} + 13h^2 V_{N-1} + 30}{15}; \\ (b_{N-1,N-3}, b_{N-1,N-2}, b_{N-1,N-1}, b_{N-1,N}) &= (E_1, E_2, E_3, E_4), \end{aligned} \quad (45)$$

where

$$\begin{aligned} E_1 &= \frac{9 - 14ah^2(V_{N-2} + V_{N-4})}{120}, \\ E_2 &= \frac{7ah^2(V_{N-2} + V_{N-3}) + 13}{15}, \\ E_3 &= \frac{7(1 - 12ah^2 V_{N-2})}{60} \pm \frac{9 - 28ah^2 V_{N-2}}{120}, \\ E_4 &= \frac{7ah^2(V_{N-2} + V_{N-1}) + 13}{15}; \\ (c_{N-1,N-3}, c_{N-1,N-2}, c_{N-1,N-1}, c_{N-1,N}) &= (-\frac{7}{60}a, \frac{7}{15}a, -\frac{7}{10}a \pm \frac{7}{60}a, \frac{7}{15}a) \end{aligned} \quad (46)$$

with the plus sign for even-parity wave functions and minus sign for odd parity wave functions.

$$(a_{N,N-2}, a_{N,N-1}, a_{N,N}) = \begin{cases} (D_{1,3}, D_{1,2}, D_{1,1e}) & \text{for even parity solutions,} \\ (0, 0, D_{1,1o}) & \text{for odd parity solutions,} \end{cases} \quad (47)$$

where

$$\begin{aligned} D_{1,1e} &= \frac{-9h^2 V_{N-3} + 14ah^4 V_{N-1} V_{N-3} + 120}{60}, \\ D_{1,2} &= -2\frac{7ah^4 V_{N-1} V_{N-2} + 13h^2 V_{N-2} + 30}{15}, \\ D_{1,3} &= \frac{120 - 7h^2 V_{N-1} + 42ah^4 V_{N-1}^2}{60}, \\ D_{1,1o} &= \frac{120 - 7h^2 V_{N-1} + 42ah^4 V_{N-1}^2}{60}. \end{aligned}$$

$$(b_{N,N-2}, b_{N,N-1}, b_{N,N}) = \begin{cases} (M_{1,3}, M_{1,2}, M_{1,1e}) & \text{for even parity solutions,} \\ (0, 0, M_{1,1o}) & \text{for odd parity solutions,} \end{cases} \quad (48)$$

where

$$\begin{aligned} M_{1,1e} &= \frac{9 - 14ah^2(V_{N-1} + V_{N-3})}{60}, \\ M_{1,2} &= 2 \frac{7ah^2(V_{N-1} + V_{N-2}) + 13}{15}, \\ M_{1,3} &= \frac{7h^2(1 - 12ah^2V_{N-1})}{60}, \\ M_{1,1o} &= \frac{7h^2(1 - 12ah^2V_{N-1})}{60} : \\ (c_{N,N-2}, c_{N,N-1}, c_{N,N}) &= \begin{cases} (-\frac{7}{30}a, \frac{14}{15}a, -\frac{7}{10}a) & \text{for even parity solutions} \\ (0, 0, -\frac{7}{10}a) & \text{for odd parity solutions} \end{cases} \end{aligned} \quad (49)$$

all other matrix elements of A, B and C are equal to zero and $Y = (y_0, y_1, \dots, y_{N-1})^T$.

Since C is a non-singular matrix, the problem (34) is equivalent to the problem

$$(h^2 E^{\text{calc}})^2 Y - h^2 E^{\text{calc}} C^{-1} B Y - C^{-1} A Y = 0. \quad (50)$$

By introducing the variable $W = h^2 E^{\text{calc}} Y$ the above equation can be transformed into the standard eigenvalue problem

$$\begin{bmatrix} C^{-1}B & C^{-1}A \\ I & 0 \end{bmatrix} \begin{bmatrix} W \\ Y \end{bmatrix} = h^2 E^{\text{calc}} \begin{bmatrix} W \\ Y \end{bmatrix}. \quad (51)$$

It is easy to see that the matrices A, B, C and

$$\begin{bmatrix} C^{-1}B & C^{-1}A \\ I & 0 \end{bmatrix}$$

are not symmetric. Transforming the matrix

$$\begin{bmatrix} C^{-1}B & C^{-1}A \\ I & 0 \end{bmatrix}$$

to a Hessenberg form, its eigenvalues can be obtained using, for example, the well known QR method.

4.2. Numerical results

To test the validity of the proposed new method we have applied the new method to the potentials (2)–(4) for specific choices of the parameters, for the first case. Based on the work of Fack and Vanden Berghe [15], we have chosen the appropriate values of R which are shown in tables. For the second case we have applied the new method to the potentials (5)–(6). For comparison purposes we have used the extended Numerov method of Fack and Vanden Berghe [16].

In Table 1 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E_n , $n = 1, \dots, 4$ for the potential (2) with $\lambda = \gamma = 0$ using the new method developed in Section 3 and the extended Numerov method of Fack and Vanden Berghe [16], for $h = 0.1$.

In Table 2 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E_n , $n = 1, \dots, 4$ for the potential (2) with $\lambda = \gamma = 0.1$ using the new method developed in Section 3 and the extended Numerov method of Fack and Vanden Berghe [16], for $h = 0.1$.

In Table 3 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E for the potential (2) and for various choices of λ and γ using the new method developed in Section 3 (which is presented as method [b]) and the extended Numerov method of Fack and Vanden Berghe [16] (which is presented as method [a]). $R = 10$ and $h = 0.1$. By $(j+)$ we present the j th even eigenvalue, while by $(j-)$ we present the j th odd eigenvalue.

In Table 4 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E for the potential (3) and for various choices of β using the new method developed in Section 3 (which is presented as method [b]) and the extended Numerov method of Fack and Vanden Berghe [16] (which is presented as method [a]). $R = 4$ and $h = 0.04$.

In Table 5 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E for the potential (4) and for various choices of n and m using the new method developed in Section 3

Table 1

Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the potential (2) with $\lambda = \gamma = 0$, produced by the extended Numerov method of Fack and Vanden Berghe [16] and the present method for $h = 0.1$. We note that $R = 10$ (see [15]). Total real time of computation (in s) for $h = 0.1$

Exact eigenvalues E_n	Extended Numerov method ([16])	Present method
1	$3.4 \cdot 10^{-9}$	$1.3 \cdot 10^{-10}$
3	$3.0 \cdot 10^{-8}$	$1.7 \cdot 10^{-9}$
5	$1.4 \cdot 10^{-7}$	$4.4 \cdot 10^{-10}$
7	$4.3 \cdot 10^{-7}$	$1.7 \cdot 10^{-8}$
Total real time of computation (s)	3.22	3.23

Table 2

Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the potential (2) with $\lambda = \gamma = 10.0$, produced by the extended Numerov method of Fack and Vanden Berghe [16] and the present method for $h = 0.1$. We note that $R = 10$ (see [15]). We assume as exact eigenvalues the eigenvalues obtained using the extended Numerov method of Fack and Vanden Berghe [16] with $R = 10$ and $h = 0.05$ presented in the paper of Fack and Vanden Berghe [16]. Total real time of computation (in s) for $h = 0.1$

Exact eigenvalues E_n	Extended Numerov method [16]	Present method
1.580022327	$3.9 \cdot 10^{-8}$	$7.6 \cdot 10^{-9}$
3.879036830	$4.1 \cdot 10^{-8}$	$6.0 \cdot 10^{-9}$
5.832767530	$1.9 \cdot 10^{-7}$	$3.1 \cdot 10^{-8}$
7.903154152	$4.8 \cdot 10^{-7}$	$4.5 \cdot 10^{-9}$
Total real time of computation (s)	3.31	3.32

Table 3

Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the potential (2) for various choices of λ and γ , produced by the extended Numerov method of Fack and Vanden Berghe [16] (presented as [a]) and the present method (presented as [b]). $R = 10$ (see [15]) and $h = R/N = 0.1$. Total real time of computation (in s) for $h = 0.1$

$\lambda(j\pm)$	γ	Exact eigenvalues	[a]	[b]
$-0.42(1+)$	0.1	0.8^a	$1.4 \cdot 10^{-9}$	$8.7 \cdot 10^{-11}$
$-0.67 + 0.1\sqrt{3.05}(2+)$	0.1	$2.3 + \sqrt{3.05}^a$	$6.8 \cdot 10^{-8}$	$2.5 \cdot 10^{-9}$
$-0.46(1-)$	0.1	2.4^a	$1.4 \cdot 10^{-8}$	$6.3 \cdot 10^{-10}$
$-0.73 + 0.1\sqrt{4.09}(2-)$	0.1	$3.7 + \sqrt{4.09}^a$	$2.3 \cdot 10^{-7}$	$7.2 \cdot 10^{-9}$
Total real time of computation (s)			3.43	3.45

^aExact eigenvalues presented in [15].

Table 4

Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the potential (3) for various choices of β , produced by the extended Numerov method of Fack and Vanden Berghe [16] (presented as [a]) and the present method (presented as [b]). $R = 4$ (see [15]) and $h = R/N = 0.04$. For the exact eigenvalues, see [16]. Total real time of computation (in s) for $h = 0.04$

β	Parity	Exact eigenvalues	[a]	[b]
11	Even	-8	$2.3 \cdot 10^{-8}$	$2.0 \cdot 10^{-9}$
		0	$4.7 \cdot 10^{-8}$	$3.7 \cdot 10^{-9}$
		8	$2.8 \cdot 10^{-7}$	$1.7 \cdot 10^{-8}$
13	Odd	-11.313708500	$3.3 \cdot 10^{-8}$	$4.4 \cdot 10^{-9}$
		0	$1.6 \cdot 10^{-7}$	$1.2 \cdot 10^{-8}$
		11.313708500	$7.5 \cdot 10^{-7}$	$4.2 \cdot 10^{-8}$
15	Even	-15.077508510	$4.7 \cdot 10^{-8}$	$2.6 \cdot 10^{-9}$
		-3.559316943	$1.4 \cdot 10^{-7}$	$1.5 \cdot 10^{-8}$
		3.559316943	$3.4 \cdot 10^{-7}$	$2.1 \cdot 10^{-8}$
17	Odd	15.077508510	$1.8 \cdot 10^{-6}$	$9.9 \cdot 10^{-8}$
		-19.158416010	$6.3 \cdot 10^{-8}$	$2.8 \cdot 10^{-9}$
		-5.740652916	$2.8 \cdot 10^{-7}$	$2.9 \cdot 10^{-9}$
		5.740652916	$9.5 \cdot 10^{-7}$	$6.3 \cdot 10^{-8}$
		19.158416010	$3.8 \cdot 10^{-6}$	$2.1 \cdot 10^{-7}$
Total real time of computation (s)			12.49	12.51

(which is presented as method [b]) and the extended Numerov method of Fack and Vanden Berghe [16] (which is presented as method [a]). In the table, as input, we present the appropriate values of R and $h = R/N$.

In Table 6 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E for the potential (5) using the new method developed in Section 3 (which is presented as method [b]) and the extended Numerov method of Fack and Vanden Berghe [16] (which is presented as method [a]). $R = 10.0$ and $h = 0.0625$.

Table 5

Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the potential (4) for various choices of m and n produced by the extended Numerov method of Fack and Vanden Berghe [16] (presented as [a]) and the present method (presented as [b]). The values of R and $h = R/N$ are presented in the second column. For the exact eigenvalues, see [30]. Total real time of computation (in s) for h given in second column

n	m	$R(h)$	Exact eigenvalues	[a]	[b]
1	1	2.48(0.0248)	-2	$1.3 \cdot 10^{-10}$	$4.2 \cdot 10^{-11}$
			0	$4.6 \cdot 10^{-10}$	$1.3 \cdot 10^{-11}$
2	1	2.48(0.0248)	$-2(1 + \sqrt{2})$	$6.2 \cdot 10^{-10}$	$7.4 \cdot 10^{-11}$
			-4	$1.1 \cdot 10^{-9}$	$2.5 \cdot 10^{-10}$
			$2(\sqrt{2} - 1)$	$2.3 \cdot 10^{-9}$	$6.8 \cdot 10^{-11}$
1	10	1.32(0.0132)	-11	$6.6 \cdot 10^{-9}$	$5.7 \cdot 10^{-10}$
			9	$8.5 \cdot 10^{-8}$	$2.3 \cdot 10^{-9}$
2	10	1.32(0.0132)	$-2(1 + \sqrt{101})$	$1.8 \cdot 10^{-8}$	$3.2 \cdot 10^{-10}$
			-4	$2.4 \cdot 10^{-7}$	$2.2 \cdot 10^{-9}$
			$2(\sqrt{101} - 1)$	$2.3 \cdot 10^{-7}$	$1.5 \cdot 10^{-8}$
Total real time of computation (s)				8.33	8.34

Table 6

Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ in 10^{-6} units for the potential (5) produced by the extended Numerov method of Fack and Vanden Berghe [16] (presented as [a]) and the present method (presented as [b]) for $h=0.0625$; $R=10$ (see [31]). For the exact eigenvalues, see [1, 28]. Total real time of computation (in s) for $h=0.0625$

Exact eigenvalues	[a]	[b]
$E_0 = -178.79853835$	30	0
$E_4 = -110.80857207$	4 921	49
$E_8 = -59.006564758$	25 360	225
$E_{12} = -23.392516401$	42 261	385
$E_{16} = -3.9664270051$	19 281	8771
Total real time of computation (s)	6.43	6.45

In Table 7 we present the absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ for the energy values E for the potential (6) using the new method developed in Section 3 (which is presented as method [b]) and the extended Numerov method of Fack and Vanden Berghe [16] (which is presented as method [a]). $R=15.0$ and $h=0.125$.

From the above-mentioned results, it is easy to see that the new method is much more accurate compared with the extended Numerov method of Fack and Vanden Berghe [16].

Table 7
Comparison of absolute errors $|E^{\text{calculated}} - E^{\text{exact}}|$ in 10^{-9} units for the potential (6) produced by the extended Numerov method of Fack and Vanden Berghe [16] (presented as [a]) and the present method (presented as [b]) for $h = 0.125$; $R = 15$. (see [31]). For the exact eigenvalues see [1]. Total real time of computation (in s) for $h = 0.125$

Exact eigenvalues	[a]	[b]
$E_0 = -49.457788728$	0	0
$E_4 = -41.232607772$	8315	307
$E_9 = -22.588602257$	737 060	47 876
$E_{13} = -3.9082324808$	515 5180	563 241
Total real time of computation (s)	9.67	9.69

5. Conclusions

In this paper a new approach to methods for the numerical solution of some specific Schrödinger equations is developed. This approach is based on the minimization of the phase-lag. Using this, we have given a direct formula to find the phase-lag of a symmetric four-step method and from this formula we have constructed a simple four-step method. The numerical results indicate that the new method is much more accurate than the extended Numerov method of Fack and Vanden Berghe [16] for a variety of potentials. The theoretical reason for this is that we have a method with phase-lag of order eight while the method of Fack et al. is a method with phase-lag of order six. This is the first time that the phase-lag is applied to the construction of a matrix method. We note here that for the computation of the eigenvalues in the case of the Morse potential (5) and the Woods–Saxon potential (6) there are, also, other numerical methods which give very accurate results (see [37, 20–22]). These methods are based on the shooting technique, which is completely different from the technique used by the proposed matrix method.

However, it is clear that we can obtain much more accurate results if the phase-lag can be minimised even more. This is possible if we include other layers in the method (27).

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Appendix A. Phase-lag analysis of the method of Fack et al. [16]

Applying the method of Fack et al. [16] to the scalar test equation (8) we have the difference equation (9) (for $k = 2$) with the associated characteristic equation (10) with

$$A_2(H) = -\frac{1}{240}H^2,$$

$$\begin{aligned} A_1(H) &= 1 + \frac{1}{10}H^2, \\ A_0(H) &= -2 + \frac{97}{120}H^2. \end{aligned} \quad (\text{A.1})$$

If we apply the method of Fack et al. [16] with characteristic equation given by (10)–(13), where $A_i(H), i = 0(1)2$ are given by (A.1) and expanding $\cos(2H)$ and $\cos(H)$ via Taylor series, we have the following expression for the phase-lag:

$$t = \frac{31H^7}{120960} + O(H^9). \quad (\text{A.2})$$

So, the method of Fack et al. [16] is a sixth-order method with phase-lag of order six.

Appendix B. Interval of periodicity

Consider a symmetric four-step method for the numerical solution of (7). Application of this method to the scalar test equation (8) leads to the difference equation (9) with $k = 2$ which is associated with the characteristic equation (10).

Substituting in (10) (with $k = 2$) $s = (1 + z)/(1 - z)$ we have that

$$(2A_2(H) - 2A_1(H) + A_0(H))z^4 + 2(6A_2(H) - A_0(H))z^2 + 2A_2(H) + 2A_1(H) + A_0(H) = 0. \quad (\text{B.1})$$

From (B.1) we have the polynomials

$$\begin{aligned} P_1(H) &= 2A_2(H) - 2A_1(H) + A_0(H), \quad P_2(H) = 12A_2(H) - 2A_0(H), \\ P_3(H) &= 2A_2(H) + 2A_1(H) + A_0(H). \end{aligned} \quad (\text{B.2})$$

Theorem B.1. *All symmetric four-step methods have a nonempty interval of periodicity $(0, H_0^2)$ if, for all $H \in [0, H_0]$,*

$$P_i(H) \geq 0, \quad i = 1(1)3, \quad \text{and} \quad N(H) = P_2(H)^2 - 4P_1(H)P_3(H) \geq 0. \quad (\text{B.3})$$

Proof. From Definition 1 and with $s = \exp(i\theta)$ we obtain

$$z = i \tan(\theta/2), \quad z^2 \leq 0. \quad (\text{B.4})$$

If $P_1(H) \neq 0$, (B.1) will vanish for

$$z^2 = \frac{-P_2(H) \pm (P_2(H)^2 - 4P_1(H)P_3(H))^{1/2}}{2P_1(H)} \quad (\text{B.5})$$

and from (B.4) we obtain the conditions (B.3). $\square$

Based on Appendix A we find that the extended Numerov method of Fack et al. [14] has interval of periodicity $(0, 6.7)$. From (29) we find that our new method has interval of periodicity $(0, 7.3)$.

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